OGC Builder Days Code Sprint May 2026

Insights & Challenges

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Since June 2025 conducting research on what seems right, correct, fun, or appropriate.

Covering

- Metadata architecture and history
- Geospatial Standards
- Psychology
- Arts
- Software Engineering
- Logics
- Games design
- Mathematics

Collaborating with academia and industry experts in

- Gen and agentic GeoAI
- Remote sensing
- Philosophy
- Libraries
- Museums, Cultural Heritage
- Archives
- Energy

Going back to this code sprint

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Everyone wants a graph ontology now

Why?

They:

- Build relationships
- Enable reasoning
- Enable interoperability
- Support extensibility [*1]

And of course

AI

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But:

- "It will take years" [*1]
- "There is a lot of work to do" [*2]
- Geospatial information in other domains (space/location) [*3]
- Other domains information in Geospatial systems (blob) [*3]
- Metadata is still "something to be added" [*4]

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For ontology-based operable systems we need:

- Reduce (a lot) the barrier for domain experts [*5]
- Consciously balance rigor with usability
- Metadata as the core of the system, not an addendum [*4]
- Understand that the rise of complexity takes us closer to philosophy [*6]
- Have federated application profiles across all domains in the system
- Move "from isolated metadata to a verifiable Claims Ecosystem" (IPT) [*7]
- Know when we do not need a graph [*8]

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Interoperate geospatial information with the application domain [*3]

IES4 *spacetime* mereology seems like an excellent job towards this. [*9]

"The Many and the One" [*6] problem becomes fundamental and nuanced,

entities are no longer defined only as parts of memberships, but also by dynamic relational structures.

An entity's -- or entities' -- meaning or identity may be fully or partially altered by their spatiotemporal values.

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Challenge: metadata as something to add [*4]

A generic “metadata” property or container might suggest

that important contextual distinctions were not part

of the core model but were instead

appended later as an undifferentiated layer.

At minimum, the name should reflect what kind of context it actually contains.

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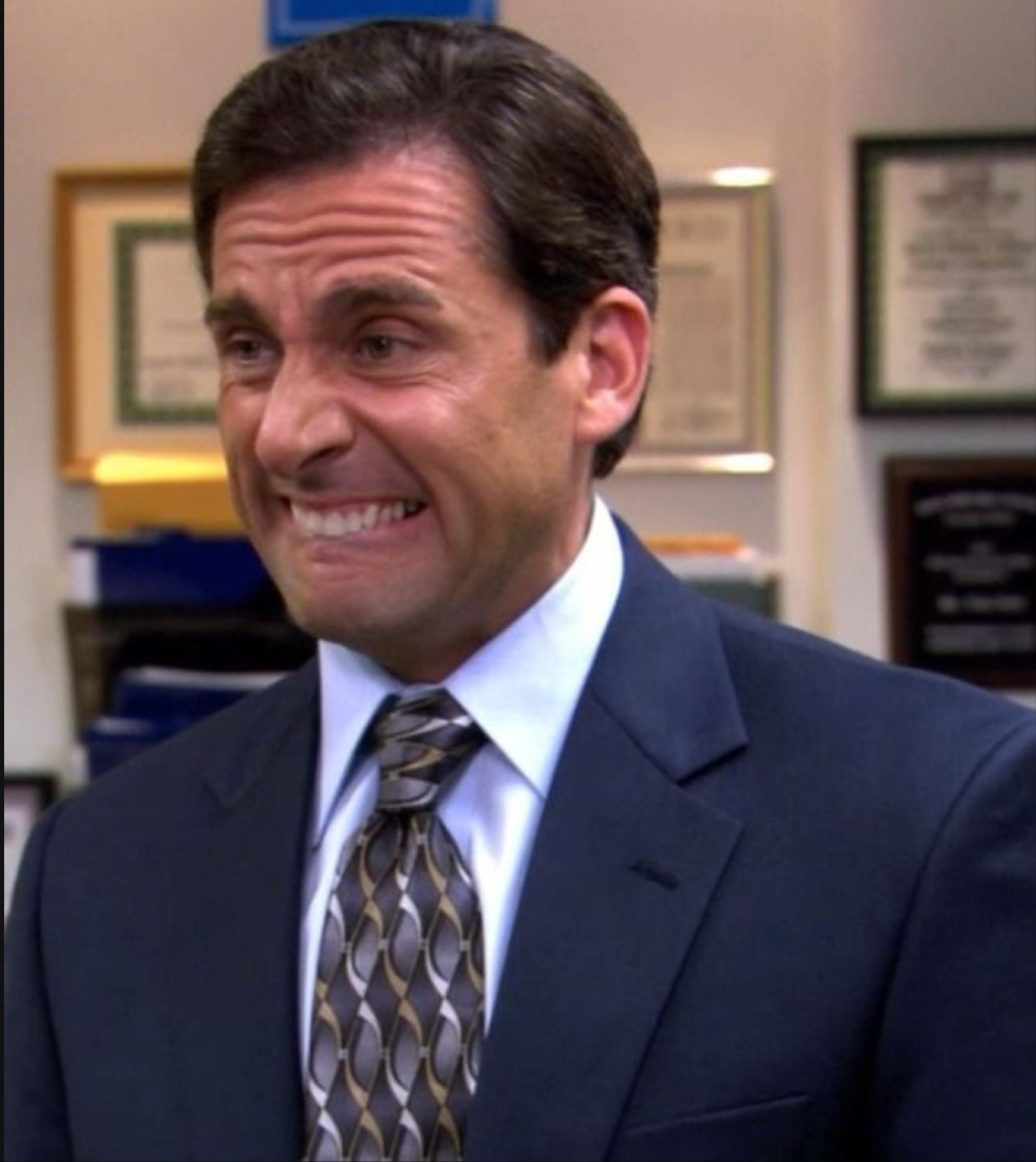
Editing graphs is not very human [*5]

```
ex:exposicion_meninas_madrid
    a crm:E5_Event ;
    crm:P2_has_type <http://vocab.getty.edu/aat/30005476
    crm:P4_has_time-span [
        a crm:E52_Time-Span ;
        crm:P82a_begin_of_the_begin "2024-05-01"
    ] ;
    crm:P7_took_place_at [
        a crm:E53_Place ;
        rdfs:label "Museo del Prado"
    ] ;
    crm:P1_is_identified_by [
        a crm:E33_E41_Linguistic_Appellation ;
        crm:P2_has_type <http://vocab.getty.edu/aat/3004
        rdfs:label "Exposición de las Meninas en Madrid"
    ] ;
    crm:P12_occurred_in_the_presence_of ex:meninas ;
    crm:P14_carried_out_by [
        a crm:E39_Actor ;
        rdfs:label ex:velazquez_diego
    ] .

ex:meninas
```



Mapping definitions [*5]

Allow us to make tree or tabular projections of a knowledge graph to:

- Hide complexity at granular levels of abstraction
- Translate properties and descriptions into the language of domain experts

Properties

class	exhibition
nombre	Exposición de las Meninas en Madrid
fecha_inicio	01/05/2024
lugar	Museo del Prado
organizador	Ministerio de Cultura de España
objetos	meninas

+ Add property

Alejandro is working on this in the building blocks / naming authority. [*2]

Challenge: when not to use a graph [*8]

If a system can be represented as a set-based model without hidden

or informal distinctions that matter for reasoning

an explicit ontology layer is unnecessary.

Otherwise, the model is either underspecified or implicitly

relying on structure that should be made explicit.

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A bright AI future vision

Federated Ontologies
+ i18n definitions
+ Different abstraction levels / interfaces
= Enable domain experts to produce and improve knowledge graphs
⇒ Explainable, Trustworthy Neuro-symbolic AI Systems
⇒ Improve human understanding, not just provide answers
⇒ Break epistemological silos between knowledge domains

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To build this, we need people with:

- Information architecture background
- Expertise across different domains
- Passion for knowledge, rigor, and psychological flexibility
- Focus on both robots and humans
- Interest in making stronger links between academia and industry

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Which aligns with

the kind of interdisciplinary work I want to help build

- Information architecture background
- Expertise across different domains and roles
- Passion for knowledge, rigor, and psychological flexibility
- Focus on building for both robots and humans
- Making stronger links between academia and industry

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References

- [∗1] Tutorial Introducing GIMI – Joe Stufflebeam (Teknicare) & Devon Sookhoo (Teknicare)
- [∗2] Conversation with Alejandro Villar and Joana Simoes during the code sprint
- [∗6] Digging into the upper ontologies presented in [∗9], I found that IES is based on BORO + Plural logic from the book "The Many and the One, A Philosophical Study of Plural Logic"
- [∗7] Data Quality for Integrity, Provenance, Trust – Andreas Matheus (Secure Dimensions), Lucio Colaiacomo (4113 Engineering), & Joan Masó (UAB - CREAF)
- [∗9] GeoSPARQL and the UK Information Exchange Standard (IES) - Paul Cripps (Dstl)
- [∗3], [∗4], [∗5], and [∗8] serve as semantic clustering among the slides

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